

Prior Trial Effects on Working Memory in Schizophrenia, Bipolar Disorder, and Healthy Control Individuals

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 Supplemental content

IMPORTANCE Healthy individuals typically show attractive serial dependence in working memory tasks, with responses biased toward recent inputs, whereas people with schizophrenia (SZ) reliably show the opposite—repulsive biases. Determining whether individuals with bipolar disorder (BD) and a history of psychosis exhibit similar or distinct serial-dependence patterns is essential for clarifying shared vs disorder-specific cognitive mechanisms across the psychosis spectrum.

OBJECTIVE To determine whether people with BD exhibit the repulsive bias seen in people with SZ or the attractive bias seen in healthy control (HC) individuals.

DESIGN, SETTING, AND PARTICIPANTS This case-control study included individuals who met *DSM-5* criteria for SZ or BD with a history of psychosis, along with matched HC participants recruited across 5 US sites. Clinical samples were recruited from outpatient psychiatric clinics and day programs, and HC participants were recruited from the community. A working memory task was used in which participants remembered the orientation of a teardrop-shaped object and reproduced it using a computer mouse after a variable delay. The Brief Psychiatric Rating Scale was used to quantify symptom severity. The study was conducted from September 2023 to April 2025.

MAIN OUTCOMES AND MEASURES The primary outcome measure was the bias index, used to evaluate the extent to which current-trial responses were biased toward or away from previous-trial targets.

EXPOSURES Diagnostic group and the orientation of the previous trial's target during a spatial working memory paradigm.

RESULTS A total of 41 participants were included in BD group (21 [51%] male; mean [SD] age, 37.02 [11.53] years), 30 in the SZ group (18 [60%] male; mean [SD] age, 37.5 [9.04] years), and 27 in the HC group (16 [59%] male; mean [SD] age, 39.98 [11.01] years). Overall bias differed across groups, with people with SZ (mean, -1.99 ; 95% CI, -2.74 to 1.23) demonstrating significantly greater repulsive bias than both people with BD (mean, 0.44 ; 95% CI, -0.25 to 1.13) and HC individuals (mean, 1.82 ; 95% CI, 1.31 to 2.32). Fifteen of 27 HC individuals exhibited attractive bias, with none showing repulsive bias, whereas 17 of 30 people with SZ exhibited repulsive bias, with none showing attractive bias. Attractive bias was common in people with BD (13 of 41), but some exhibited a repulsive bias (7 of 41).

CONCLUSIONS AND RELEVANCE In this case-control study, attractive and repulsive serial biases differed between HC individuals and people with SZ, with some people with BD showing the HC pattern and few showing the SZ pattern. These results suggest that serial bias may serve as a valuable biomarker for pathophysiology and precision psychiatry.

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Cognitive impairment is a core feature of pathology observed in people with schizophrenia (SZ),¹ with recent research showing that cognitive impairment is also characteristic of people with bipolar disorder (BD) and a history of psychosis.²⁻⁴ Although people with SZ typically exhibit more severe impairments, several studies suggest that both disorders share a broadly similar cognitive profile, differing primarily in degree rather than kind.⁵⁻⁷ This quantitative perspective dominates literature, raising the question of whether qualitative differences exist in specific cognitive functions and mechanisms between these disorders.

To probe this possibility, we examined serial dependence—a working memory phenomenon in which responses on the current trial are systematically biased toward (attracted toward) features of the previous trial. Healthy control (HC) individuals consistently show attraction; for instance, when remembering a target at 3 o'clock after having seen one at 1 o'clock, the recalled response is shifted toward 2:30 (Figure 1). This attraction occurs across various stimulus dimensions (eg, colors, spatial locations, and orientations)⁸ and is strongest when the current-trial and previous-trial stimuli are similar. Across multiple studies,⁹⁻¹¹ people with SZ show the opposite pattern: systematic repulsion (Figure 1) from the previous trial stimulus, which increases with longer working memory delays.^{12,13}

Although small in magnitude, these biases offer insight into computational mechanisms supporting working memory, which is associated with overall cognitive function.¹⁴⁻¹⁶ The bias direction reflects summed consequences of concurrent attractive and repulsive mechanisms, with attractive effects outweighing repulsive effects in neurotypical individuals.^{17,18} A leading model¹⁹ proposes that attraction is caused by short-term plasticity within prefrontal recurrent networks, which transiently upweights synaptic connections, leading to a progressive

Key Points

Question Does the influence of prior trials on spatial working memory performance differ qualitatively in individuals with schizophrenia compared with individuals with bipolar disorder with a history of psychosis and healthy control individuals?

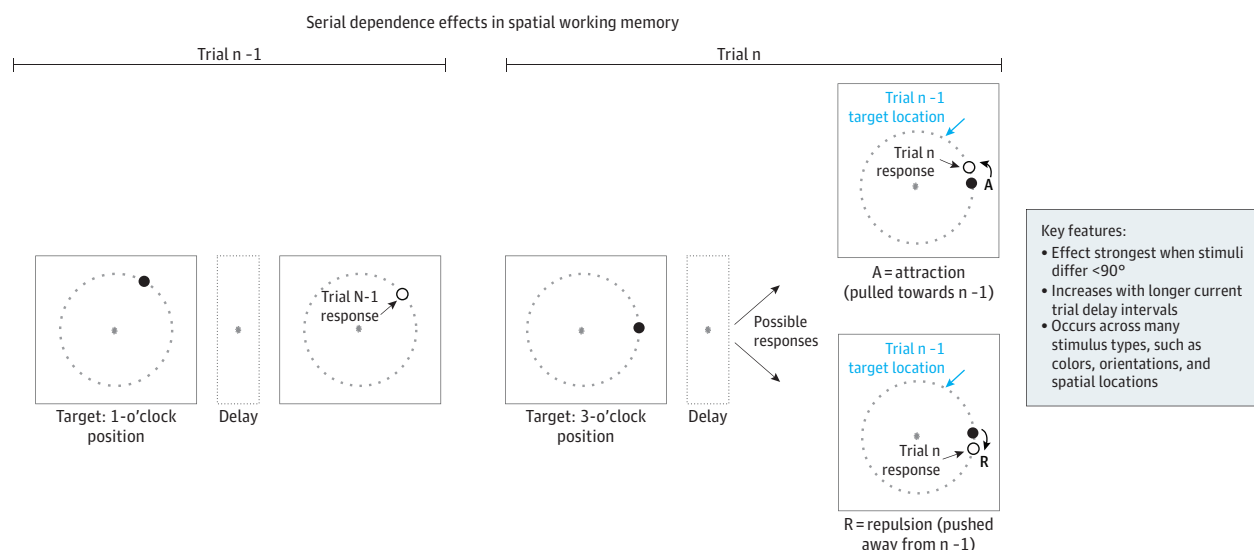
Findings This case-control study including 98 individuals found that most healthy control individuals demonstrated an attractive bias, meaning their responses were systematically pulled toward the stimulus seen on the previous trial; most people with schizophrenia showed a repulsive bias, with responses shifted away from the previous-trial stimulus, and none showed an attractive bias; and many people with bipolar disorder exhibited the attractive bias seen in healthy control individuals, although a small minority exhibited the repulsive pattern seen in people with schizophrenia.

Meaning The results suggest that serial bias may serve as a valuable biomarker for pathophysiology and precision psychiatry.

shift of the representation toward the previous-trial stimulus as the memory cycles through the recurrent network. Repulsion is thought to stem from neuronal adaptation in the visual cortex,^{20,21} where prolonged exposure to a stimulus reduces neuronal responsiveness to that feature. Working memory representations cycle through visual cortex,²²⁻²⁴ so repulsion should increase as the delay increases in working memory tasks.

In this framework, short-term plasticity dominates over adaptation in HC individuals, producing net attraction. In people with SZ, however, the balance is shifted toward adaptation, yielding net repulsion. This could result from exaggerated adaptation, consistent with increased tilt-aftereffects reported in people with SZ,^{25,26} or weakened short-term plasticity, consistent with evidence associating serial dependence

Figure 1. Illustration Showing Serial Dependence in Spatial Working Memory: Attraction and Repulsion Effects



Serial dependence effects in a delayed spatial working memory task illustrated using clock face stimuli. Trial n-1 presents a target at 1 o'clock, followed by a delay and response. In trial n, the target appears at 3 o'clock. Due to serial

effects, the response on trial n may be biased toward the prior target or away from it.

Table. Participant Characteristics

Characteristic	HCIs (n = 27)	Individuals with BD (n = 41)	Individuals with SZ (n = 30)	Statistic	P value ^a
Age, mean (SD), y	39.98 (11.01)	37.02 (11.53)	37.5 (9.04)	$F_{2,95} = .68$	.51
Sex, No. (%)					
Male	16 (59.3)	21 (51.2)	18 (60.0)	$\chi^2_2 = 1.48$	.48
Female	10 (37.0)	18 (43.9)	12 (40.0)		
Nonbinary	1 (3.7)	2 (4.9)	0		
Race, No. (%) ^b					
Black	11 (40.7)	15 (36.6)	12 (40.0)	$\chi^2_2 = 15.12$	.52
White	11 (40.7)	24 (58.5)	14 (46.7)		
Other ^c	5 (18.5)	2 (4.9)	4 (13.3)		
Education, mean (SD), y	16.79 (2.74)	14.78 (2.23)	13.91 (2.58)	$F_{2,95} = 9.28$	<.001 ^d
Mother's education, mean (SD), y	16.21 (1.99)	14.07 (3.27)	14.25 (2.65)	$F_{2,95} = 5.38$	<.001 ^d
Father's education, mean (SD), y	14.59 (2.89)	15.11 (3.02)	13.84 (3.88)	$F_{2,95} = 0.98$	.38
BPRS score, mean (SD)					
Total	NA	31.92 (6.35)	40.04 (8.37)	$t_{69} = -3.98$	<.001
Positive symptoms	NA	4.56 (1.36)	8 (4.39)	$t_{69} = -3.72$	<.001
Negative symptoms	NA	2.17 (1.75)	4.54 (2.64)	$t_{69} = -3.84$	<.001
Disorganization	NA	1.97 (0.84)	2.33 (1.34)	$t_{69} = -1.17$	.25
Depression	NA	3.22 (2.14)	3.54 (2.17)	$t_{69} = -0.57$	.57
Mania	NA	2.61 (1.84)	2.75 (2.09)	$t_{69} = -0.09$	.93
CAINS MAP	NA	8.58 (6.62)	13.2 (7.94)	$t_{69} = -1.91$	.07
CAINS EXP	NA	1.88 (1.73)	3.4 (2.47)	$t_{69} = -1.92$	.07
Antipsychotic medication, total CPZ, mean (SD), mg/d	NA	173.47 (112.22)	479.48 (431.72)	$t_{69} = -2.90$	.008
Psychotropic drugs other than antipsychotics, No. (%) ^a					
Mood stabilizers	NA	23 (56.1)	7 (23.3)	NA	NA
Antidepressant	NA	13 (31.7)	10 (33.3)	NA	NA
Anxiolytic	NA	3 (7.3)	9 (30)	NA	NA
Stimulant	NA	4 (9.8)	3 (10.0)	NA	NA
Other	NA	1 (2.4)	4 (13.3)	NA	NA

Abbreviations: BD, bipolar disorder and history of psychosis; BPRS, Brief Psychiatric Rating Scale; CAINS, Clinical Assessment Interview for Negative Symptoms; CPZ, chlorpromazine; EXP, expression subscale; HCIs, healthy control individuals; MAP, motivation and pleasure subscale; NA, not applicable; SZ, schizophrenia.

^a One-way ANOVA, *t* tests, and χ^2 tests were used to analyze group differences.

^b Race and ethnicity were obtained via participant self-report using standardized, multiple-choice categories consistent with National Institutes of Health reporting guidelines. These variables were collected to ensure complete characterization of the study sample and to support assessment of generalizability.

^c The other category included participants who self-identified as Asian, Native American, or multiple races or whose race and ethnicity were unknown. These groups were consolidated due to small cell sizes, which limited the stability and interpretability of group-specific estimates.

^d HI > SZ/BD; SZ = BD.

with N-methyl-D-aspartate (NMDA) receptor-dependent dynamics disrupted in SZ.¹⁰ Prior work²⁶ indicates that adaptation-driven tilt-aftereffects are not exaggerated in people with BD, raising the possibility that serial dependence patterns could differentiate people with SZ and people with BD, offering a mechanistically meaningful biomarker across the psychosis spectrum.

Thus, the objective of this study was to examine whether people with BD exhibit SZ-like repulsive pattern or HC-like attractive pattern. A qualitative divergence would indicate important biological distinctions, whereas similarity would suggest shared neural mechanisms.

Methods

Participants

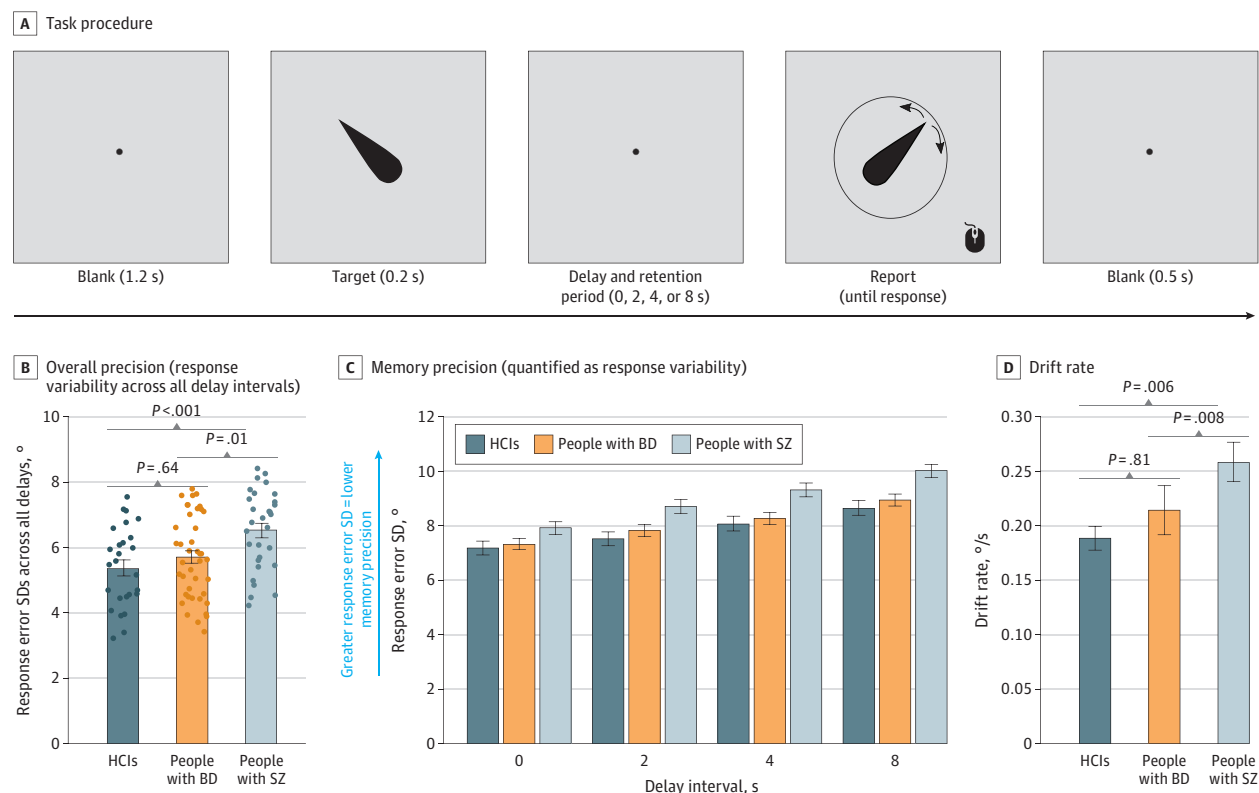
In this case-control study, we recruited clinically stable individuals meeting *DSM-5* criteria²⁷ for people with SZ (n = 30) and people with BD with a history of psychosis (n = 41), along with 27 HC individuals, recruited via internet and print advertisements and from outpatient clinics across 5 academic sites in Baltimore Maryland; Chicago, Illinois;

St Louis, Missouri; Irvine, California; and Minneapolis, Minnesota (see eFigure 2 in Supplement 1, which also shows that site was not associated with performance). HC individuals had no psychiatric, neurological, or cognitive impairments and no first-degree relatives with psychosis. Procedures were approved by a central institutional review board and followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.²⁸ Individuals gave written informed consent prior to study participation. Symptoms were assessed using the Brief Psychiatric Rating Scale.²⁹ Participant characteristics are summarized in the Table. The study was conducted from September 2023 to April 2025.

Stimuli and Procedure

Participants completed a working memory task developed previously¹⁰ (Figure 2A) in which they viewed a teardrop-shaped stimulus for 200 ms, remembered its orientation across a delay of 0, 2, 4, or 8 seconds, and reproduced the orientation using a mouse. Stimuli were selected from 12 equally spaced orientations. After 16 practice trials, participants completed 288 trials across 3 blocks. Further details are described in eMethods 1 in Supplement 1.

Figure 2. Illustration and Bar Charts Showing Task Design and Group Differences in Memory Precision



A, Each trial began with a 1.2-second blank screen, followed by a 0.2-second presentation of a target stimulus (a teardrop-shaped object). After a variable delay interval (0, 2, 4, or 8 seconds), participants were prompted to reproduce the orientation of the target using the mouse. A 0.5-second blank screen followed the response before the next trial. B, Higher standard deviation (SD) indicates lower memory precision. C, To compare overall precision among groups, we calculated overall precision by averaging the standard deviation of response error across delay intervals. Averaged across delay intervals,

individuals with schizophrenia (SZ) exhibited significantly reduced memory precision compared to healthy control individuals (HCIs) and those with bipolar disorder and history of psychosis (BD), while results for individuals with BD did not significantly differ from those of HCIs. D, Drift rate (degrees/s), representing the rate of decline in memory precision over time, is shown for each group. Statistical comparisons indicate differences in the dynamics of memory degradation. Error bars represent standard error of the mean.

Statistical Analysis

Our key measure was the signed response error (eFigure 1 in Supplement 1)—the difference between reported orientation and true orientation for the current trial, with the sign indicating direction relative to the prior trial. Based on prior work,^{9,10} our principal analysis averaged across trials with small orientation differences (30° and 60°) and long delays (4 and 8 seconds), yielding an overall bias score. Group differences in overall bias were tested with a 1-way analysis of variance (ANOVA), supplemented by a full 3-factor ANOVA on the signed response error, examining group, delay, and orientation difference. Each participant's bias was also classified as attractive, repulsive, or indeterminate using standard error estimates.

Working memory precision was assessed using standard deviation of unsigned errors across trials for each delay interval, and degradation over time was quantified using drift rate (slope of the function relating the standard deviation to delay duration). Correlations between bias and symptoms, cognition, or medication were also assessed. All statistical tests were 2-tailed ($\alpha = .05$). Pairwise comparisons were evaluated using

Tukey-adjusted tests. Additional details are provided in eMethods 2 in Supplement 1.

Results

Working Memory Precision

We first examined working memory precision, asking whether working memory representations showed reduced precision (greater spatial scatter) in the clinical groups compared to HC individuals, and whether precision decayed more rapidly in clinical groups. Precision was averaged across orientations to isolate memory degradation effects independent of angular spacing. People with SZ showed significantly poorer working memory precision than both people with BD and HC individuals, who did not differ from each other (Figure 2B; main effect of group: $F_{2,94} = 6.20$; $P = .003$; partial $\eta^2 = 0.12$). All groups exhibited decreasing precision as delay increased, but people with SZ showed a steeper decline (Figure 2C; main effect of delay duration: $F_{2,27,213,60} = 303.32$; $P < .001$; partial $\eta^2 = 0.76$;

delay $\times$ group interaction: $F_{4.54,213.60} = 4.29$; $P = .001$; partial $\eta^2 = 0.08$; 95% CI, 0.03-0.13). Drift rate analyses confirmed that people with SZ had significantly faster decay (Figure 2D) than people with BD and HC individuals ($F_{2,94} = 4.85$; $P = .01$; partial $\eta^2 = 0.09$; 95% CI, 0.01-0.21). Follow-up tests showed that people with SZ exhibited faster memory decay (0.27°/s) compared to people with BD (0.21°/s) and HC individuals (0.19°/s), who did not differ significantly.

Delay-Dependent Bias Patterns

Serial biases (bias estimation depicted in Figure 3A) are illustrated in Figure 3B. Replicating prior findings,^{9,10} we found that, when orientation differences between consecutive trials were less than 90°, HC individuals demonstrated attractive biases, whereas people with SZ exhibited pronounced repulsive biases, and these biases increased as the delay interval got longer. The novel finding was that people with BD exhibited attractive biases on average, similar to HC individuals albeit not as strong.

Quantification of Serial Bias

As in prior studies,^{9,10} we computed an overall bias score by averaging signed response errors for conditions in which serial biases are mathematically and theoretically expected to be strongest—trials with small orientation differences (30° and 60°) and long delay intervals (4 seconds and 8 seconds). Averaging across these conditions yielded a single, robust metric of serial-dependence strength for each participant, thereby maximizing statistical power to test our primary scientific hypothesis.

A 1-way ANOVA confirmed that overall bias differed across the 3 groups ($F_{2,95} = 29.38$; $P < .001$; partial $\eta^2 = 0.38$; 95% CI, 0.23-0.51). Tukey-adjusted post hoc comparisons showed that (1) people with BD demonstrated significantly weaker attractive bias than HC individuals ($t_{94} = -2.92$; $P = .01$; Cohen d , -0.72; 95% CI, -1.34 to -0.11) and (2) people with SZ demonstrated significantly greater repulsive bias than both people with BD ($t_{94} = 5.27$; $P < .001$; Cohen d , 1.27; 95% CI, 0.66 to 1.87) and HC individuals ($t_{94} = 7.51$; $P < .001$; Cohen d , 1.99; 95% CI, 1.34 to 2.64). Although sample sizes were modest, effect sizes were robust, and this difference between people with SZ and HC individuals has been observed before.⁹⁻¹¹

For completeness, we also conducted a 3-way ANOVA examining effects of group, orientation difference, and delay duration on the original signed response error scores. We obtained a significant group $\times$ orientation difference interaction ($F_{7.19,338.05} = 3.88$; $P < .001$; partial $\eta^2 = 0.08$; 95% CI, 0.03 to 0.12), reflecting differences in bias direction among groups. The temporal evolution of these group differences was captured by a significant 2-way delay $\times$ group interaction ($F_{6,282} = 4.55$; $P < .001$; partial $\eta^2 = 0.09$; 95% CI, 0.03 to 0.14). The 3-way orientation $\times$ delay $\times$ group interaction was not significant ($F_{18.45,867.19} = 0.95$; $P = .52$; partial $\eta^2 = 0.02$; 95% CI, 0.00 to 0.04), but noncrossover interactions like these have low statistical power.³⁰

Bias Categories

We used each individual's bias and standard error (SE) of estimate to classify that individual's bias as attraction (>1 SE above

0), repulsion (>1 SE below 0), or indeterminate (within 1 SE of 0). The distribution of bias patterns varied markedly across groups (Figure 3C). HC individuals predominantly exhibited attraction: 15 (55.6%) showed an attractive bias, 12 (44.4%) showed an indeterminant bias, and none showed a repulsive bias. People with SZ exhibited the opposite pattern: no participants showed an attractive bias, 12 (41.4%) showed an indeterminant bias, and 17 (58.6%) showed a repulsive bias. Among people with BD, 13 (31.7%) showed an attractive bias, 21 (51.2%) showed an indeterminant bias, and 7 (17.1%) showed a repulsive bias.

A χ^2 test revealed a strong association between group membership and bias type ($\chi^2_4 = 36.73$; $n = 97$; $P < .001$). Follow-up pairwise comparisons with Bonferroni correction showed that HC individuals and people with SZ differed significantly in the proportions of participants across the 3 classes ($\chi^2_2 = 31.97$; $n = 56$; adjusted $P < .001$). People with BD also differed significantly from people with SZ ($\chi^2_2 = 18.1$; $n = 70$; adjusted $P < .001$). HC individuals and people with BD were not significantly different, ($\chi^2_2 = 7.01$; $n = 68$; adjusted $P = .09$).

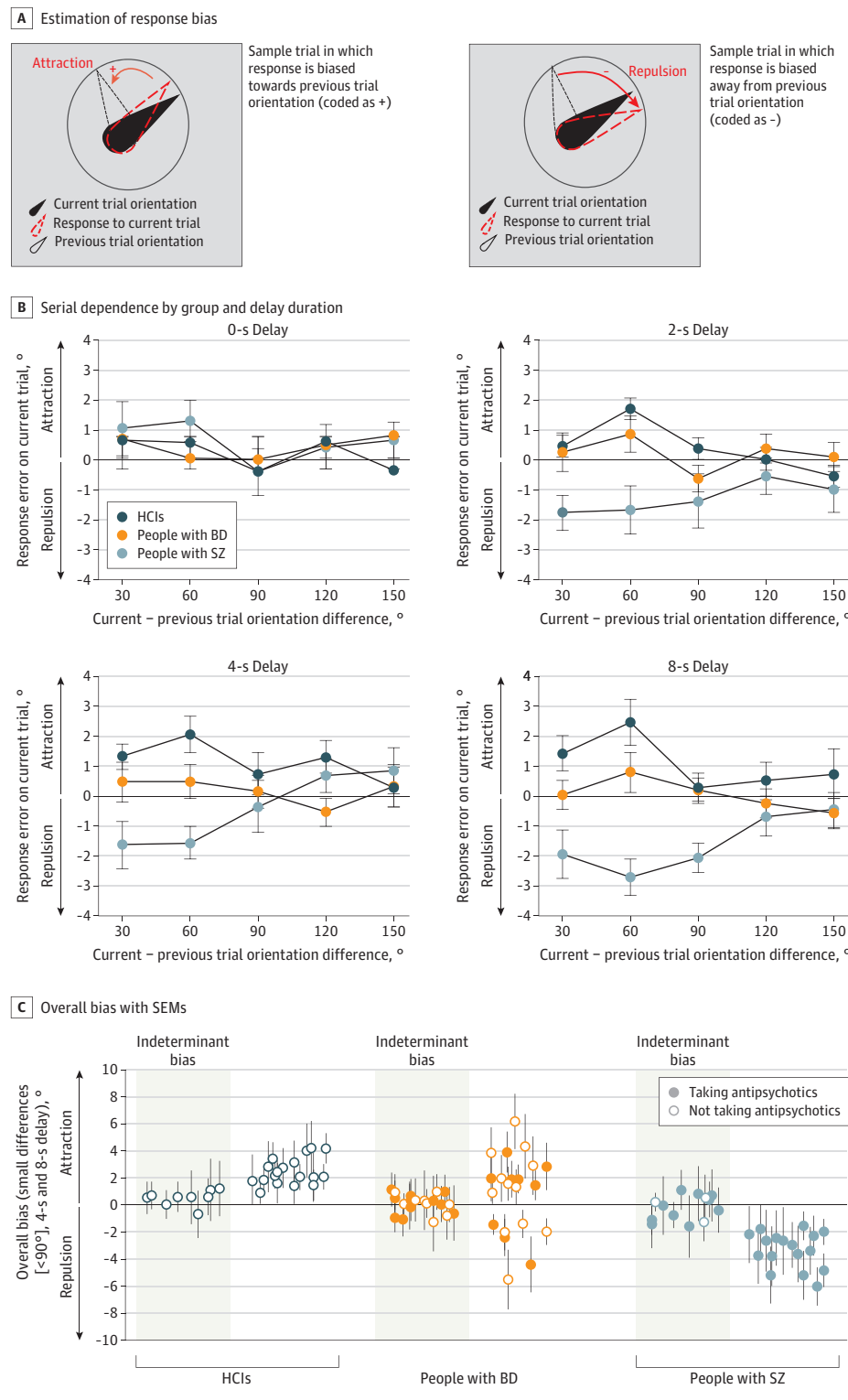
Correlations

We examined associations among the overall bias index, symptom ratings, and medication dosage in people with BD and people with SZ. As shown in Figure 4A, greater repulsion was associated with higher mania scores in people with BD ($r, -0.49$; 95% CI, -0.70 to -0.20; $P = .01$; adjusted $P = .04$). In people with SZ (Figure 4B), greater repulsion was associated with more severe positive symptoms ($r, -0.49$; 95% CI, -0.74 to -0.12; $P = .02$), though this did not survive false discovery rate correction. All other symptom correlations in both people with BD and people with SZ were nonsignificant after correction. No other clinical or demographic factors differentiated individuals who did vs did not exhibit repulsion in either the people with BD or people with SZ group (eFigure 3 in Supplement 1).

A partial correlation analysis was conducted to examine the association between overall bias and antipsychotic medication (chlorpromazine equivalent^{31,32}) across both SZ and BD groups. After partialling out effect of group, the correlation between bias and chlorpromazine equivalent was negligible (Spearman $\rho = 0.05$; 95% CI, -0.36 to 0.44; $P = .82$), providing no evidence that antipsychotic medications were associated with serial biases. The qualitative data in Figure 3C also support this conclusion: of 7 people with BD exhibiting repulsive biases, 3 were taking antipsychotics and 4 were not. In exploratory comparisons (eFigure 3 in Supplement 1), we examined whether individuals who exhibited repulsive bias differed from those who did not in education, age, premorbid IQ, or symptom severity. Although these analyses were limited by modest subgroup sizes, the only differentiating features were higher mania symptoms among people with participants and higher positive symptoms among people with SZ, paralleling the correlation results.

To assess the association between abnormal serial biases and working memory deficits, we analyzed correlations between the overall bias score and precision (scatter) across 4-second and 8-second delays in each group. People with SZ and people with BD showed negative associations between

Figure 3. Illustration and Dot Plot for Serial Bias

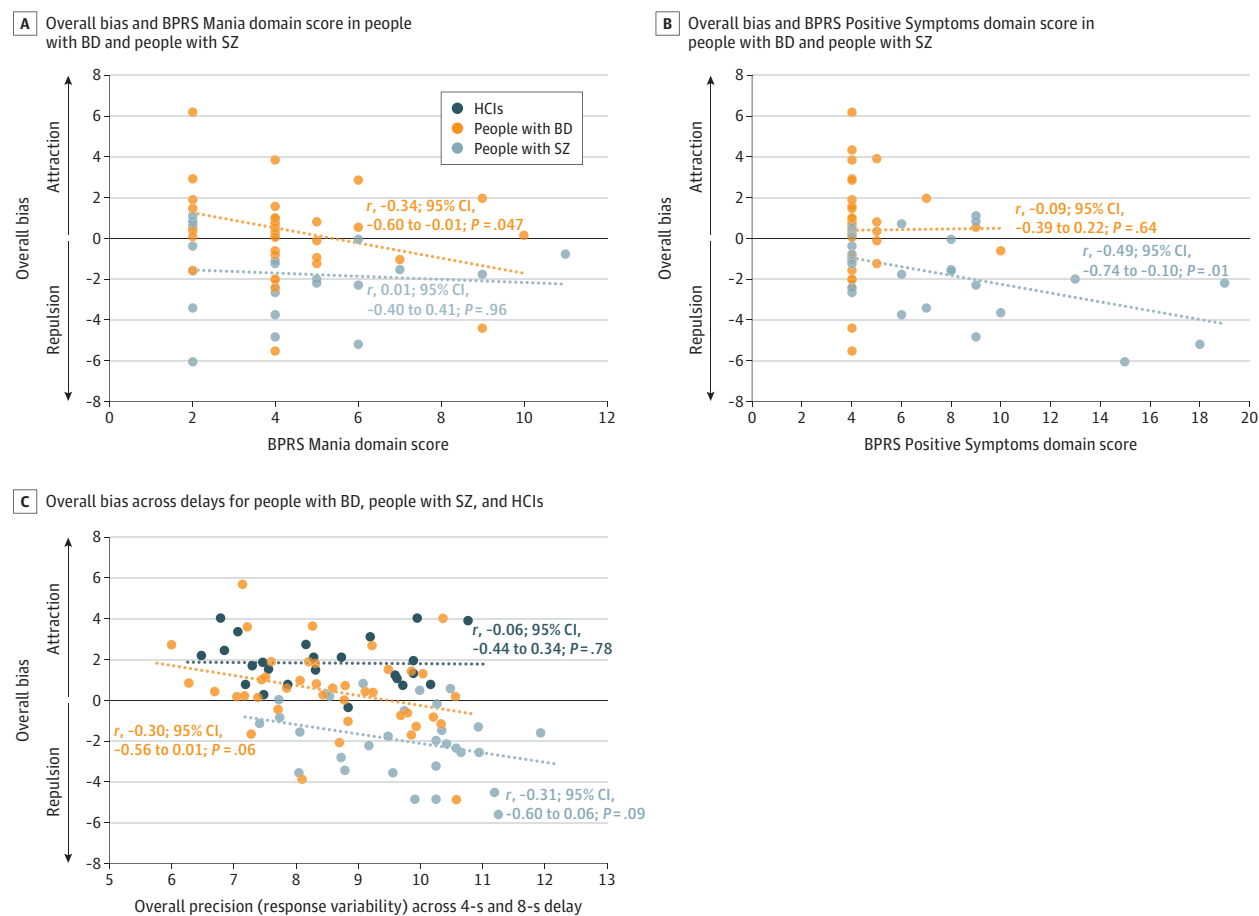


A, To determine whether there was attraction or repulsion from the previous trial orientation, the sign of the response error for the current trial orientation was designated as positive if the reported orientation matched the previous trial target and negative if it did not. If the current trial orientation was 60° and the previous trial target was -30° and a participant reported an orientation of 50° for the current trial, this would be designated as a response error of $+10^\circ$. If the participant reported an orientation of 70° , this would be signed as a response error of -10° . B, Positive response error indicates an attractive bias toward the orientation of the previous trial, while negative error indicates a repulsive bias. C, Overall bias scores were calculated from small orientation differences ($<90^\circ$) at longer delays (4 and 8 seconds). Bars indicate mean bias with standard error of the mean (SEM). BD indicates bipolar disorder; HCIs, healthy control individuals; SZ, schizophrenia.

bias and precision (people with SZ: r , -0.31 ; 95% CI, -0.60 to 0.06 ; P = .09; people with BD: r , -0.30 ; 95% CI, -0.56 to 0.01 ; P = .06), indicating that greater response scatter was associated with larger repulsive biases. HC individuals showed no association (r , -0.06 ; 95% CI, -0.44 to 0.34 ;

P = .78) (eResults 1 and eFigure 4 in Supplement 1). However, Fisher r to z tests indicated no significant differences between clinical groups and control individuals (people with SZ vs HC: z , 0.96 ; P = .17; people with BD vs HC: z , 0.93 ; P = .18). To increase power, we fit a linear regression

Figure 4. Dot Plots Showing Correlations



A, There was a significant correlation between overall bias score and the mania score in individuals with bipolar disorder and history of psychosis (BD), indicating that lower attractive bias was associated with mania scores. However, this correlation was not significant in individuals with schizophrenia (SZ). B, In individuals with SZ, greater repulsion (more negative overall bias) was

positively correlated with Brief Psychiatric Rating Scale (BPRS) positive symptoms, but this correlation was not significant in individuals with BD. Each data point represents an individual participant. C, Trends indicate higher precision may be associated with reduced attraction bias in individuals with BD and SZ, whereas no association was observed in healthy control individuals (HCIs).

predicting bias from precision, patient status (people with SZ and people with BD vs control individuals), and their interaction. The main-effects model explained significant variance ($R^2 = 0.31$; 95% CI, 0.19 to 0.45; $P < .001$), and the precision $\times$ patient interaction improved model fit ($R^2 = 0.3495$; 95% CI, 0.20 to 0.50) and was significant ($b, -0.67$; 95% CI, 1.26 to -0.16 ; $P = .047$), indicating a stronger negative precision-bias association in pooled patients than control individuals (eResults 1 and eFigure 4 in Supplement 1). In sum, although individual correlations and their comparisons were nonsignificant, the regression analysis suggests a differential bias-precision association in pooled patients relative to control individuals, warranting cautious interpretation and replication in larger samples. Further, it has previously been reported^{9,10} that repulsive serial biases in people with SZ were associated with poorer performance in standard working memory tasks. Overall, these results indicate that atypical bias may be related to working memory instability in psychosis-spectrum populations.

Discussion

Evidence from 4 independent samples,⁸⁻¹⁰ including this case-control study, shows that people with SZ exhibited a qualitative reversal of the normal attractive serial bias. One study⁹ also found that greater repulsive bias in people with SZ was correlated with poorer working memory capacity and overall cognitive ability, indicating a robust phenomenon associated with core cognitive deficits.

When we classified serial bias at the individual level, we found striking differences across groups: no HC individuals showed repulsion and most showed attraction, whereas no people with SZ showed attraction and most showed clear repulsion. Such qualitative, nonoverlapping patterns are rare and cannot be attributed to generalized poor performance or low effort. They point to a fundamental alteration in how recent experience shapes subsequent processing in people with SZ, making serial bias a promising biomarker—

any treatment that reduces repulsive bias is likely normalizing a core information-processing mechanism. People with BD showed a mixed pattern (31.7% attractive; 17.1% repulsive), markedly different from people with SZ (0% attractive; 58.6% repulsive). This diagnostic specificity further supports serial bias as a biomarker for treatment development. Early-stage interventions could target reductions in repulsive bias, which should benefit individuals who exhibit repulsion (whether people with SZ or people with BD), but would not be expected to help individuals who show attraction—consistent with a precision-medicine approach. A key future question is whether people with BD showing repulsive bias share other neural, cognitive, or symptom features with people with SZ.

In terms of precision, people with SZ exhibited impaired working memory precision compared to both people with BD and HC individuals, but people with BD did not differ from HC individuals. The lack of an impairment in people with BD relative to HC individuals might seem incompatible with prior studies showing working memory impairments in people with BD, but these have focused on measures of spatial span and working memory capacity rather than working memory precision.³³⁻³⁵

In our correlational analyses, greater repulsive bias was significantly associated with greater severity of manic symptoms in people with BD. An analogous trend was also observed in people with SZ, with greater repulsion associated with more severe psychosis, although this effect did not survive correction for multiple comparisons. Larger sample sizes or meta-analyses would be needed to have confidence in these correlations, especially given that previous studies^{9,10} have typically observed that bias is more strongly associated with cognitive ability than with symptom dimensions.

The present study contributes to an expanding body of research demonstrating that people with SZ consistently exhibit repulsive rather than attractive serial biases.⁹⁻¹¹ Potentially challenging this conclusion is a recent report from Pascucci et al³⁶ who reported an intact attractive bias in people with SZ. However, they used a 1-s delay, while prior studies have shown that differences between people with SZ and HC individuals are minimal at short delays and become clear only with delays of 3 seconds or longer.

The underlying neural cause of the repulsive bias observed in most people with SZ (and a few people with BD) remains to be determined. One candidate is reduced NMDA-dependent short-term plasticity, supported by the finding that the normal attractive serial bias is eliminated in patients with anti-NMDA receptor encephalitis.¹⁰ These patients showed no bias rather than a repulsive bias, which might suggest that impaired NMDA receptor function cannot explain the repulsion observed in people with SZ. However, this lack of bias might instead indicate that NMDA receptor function in the encephalitis patients was not sufficiently reduced to yield repulsion.

Another possible explanation for repulsion in people with SZ could be increased neuronal adaptation, which is typically associated with repulsive effects on behavior. In the tilt aftereffect, for example, a tilted adapter stimulus elicits a subsequent vertical stimulus to appear tilted in the opposite direction.^{37,38} If, for example, the adapter has an orientation of -5° from upright, neurons encoding that orientation become

less responsive, leading to greater activity of neurons coding 5° than neurons coding -5° when an upright test stimulus is presented. In the present paradigm, the stimulus from the previous trial may similarly induce adaptation, resulting in a repulsive bias in report of the target on the next trial. This repulsive bias could increase as delay increases, because the working memory representation would continue to cycle through the adapted population of neurons.

The current results do not differentiate between these 2 possibilities, but future research could investigate whether behavioral and neural indicators of adaptation strength are related to the degree of repulsion in people with SZ. Future experiments could also manipulate factors known to modulate adaptation—such as stimulus duration and contrast—and assess performance in people with SZ, HC individuals, and people with BD.

At a more conceptual level, adaptation and short-term plasticity can be considered 2 sides of the same coin. Adaptation is widely believed to increase efficiency of neural coding by focusing on changes,^{39,40} much like MPEG compression represents a video in terms of what changes from one frame to the next.⁴¹ However, this compressed representation must be decoded to control behavior,^{42,43} and attraction produced by short-term plasticity could reflect such a decoding mechanism. Both the encoding and decoding are based on a bayesian prior for environmental stability; stability means that the world can be efficiently represented by differences from moment to moment, and it also means that the decoder should tend to produce outputs that are attracted toward the past.

Both repulsion during encoding and attraction during decoding must be well calibrated with each other and with the statistics of the world. If adaptation is too strong during encoding, or if short-term plasticity is too weak during decoding, the world will be treated as being more unstable than it actually is. Thus, whether the observed repulsion is due to exaggerated adaptation or reduced short-term plasticity, it would likely impact functioning across a broad range of daily tasks and experiences. A lack of perceived environmental stability could generate excessive prediction errors, perhaps contributing to psychotic symptoms.

It is important to recognize that exaggerated adaptation and reduced short-term plasticity are not the only plausible explanations for repulsive serial biases. For example, working memory representations are less precise, less stable, and more prone to disruption by distractors in people with SZ,⁴⁴⁻⁴⁶ and repulsion could reflect a compensatory process that makes memory representations more distinctive.⁴⁷ In addition, serial biases may partially result from the response stage of the task,⁴⁸⁻⁵⁰ and disruption of dopaminergic activity in the striatum in people with SZ could impact serial biases by altering motor processes. Serial biases are an active research area that will soon offer conceptual and methodological tools to understand mechanisms behind repulsive serial bias in psychiatric populations.

Limitations

This study has limitations. We did not collect extensive cognitive measures, limiting our ability to characterize

differences among patient subgroups (ie, those with repulsive, attractive, or indeterminate biases). All people with SZ were taking medication, leaving open questions about medication-naïve patients or those at high risk. Although ANOVA analyses paralleled prior serial-dependence work and we conducted trial-level analyses with similar results (eResults 2 in [Supplement 1](#)), hierarchical or bayesian models may provide more nuanced insight with larger samples.

Conclusions

Opposite direction serial biases appear to be relatively specific to people with SZ and index a fundamental difference in how recent experience impacts working memory in people with SZ relative to HC individuals and people with BD. Serial bias measures may be attractive biomarkers to guide treatment development in SZ.

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